Lecture 07: Equity Valuation: DDMs

Questions

Choose one answer. Keep full precision until the final result. Worked solutions follow the complete question set.

1. From earnings to next year's dividend

Current EPS is $8. The firm pays out 35% of earnings and earns a sustainable ROE of 12%. Assume its payout ratio and ROE remain constant. Using $g = \text{retention} \times ROE$, what is DPS_1 ?

A. $3.1360
B. $5.6056
C. $2.8000
D. $3.0184

2. Required return with CAPM

The risk-free rate is 4%, beta is 1.2, and the expected market return is 9%. What cost of equity k should you use?

A. 10.0%
B. 14.8%
C. 9.0%
D. 6.0%

3. One-stage DDM

A mature firm just paid $\text{DPS}_0 = \$3.20.$ Dividends are expected to grow at 4% forever, and $k = 12\%$. What is the intrinsic value per share?

A. $43.26
B. $40.00
C. $41.60
D. $26.67

4. The discount rate can move the valuation

Hold next year's dividend fixed at $4 and perpetual growth fixed at 4%. If k rises from 9% to 11%, what is the percentage change in the DDM value?

A. -40.00%
B. -2.00%
C. -22.22%
D. -28.57%

5. Put a two-stage DDM together

A firm will pay $\text{DPS}_1 = \$1.50$ and $\text{DPS}_2 = \$1.80.$ From year 3 onward, dividends grow at 3% forever. The cost of equity is 10%. What is the intrinsic value per share today?

A. $29.79
B. $24.10
C. $24.74
D. $23.25

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Worked solutions

1. From earnings to next year's dividend

Answer D. $3.0184

Step 1. Separate payout from retention.

Let's start with what the firm keeps.
If it pays out 35% of earnings, the other 65% stays in the business.
We use that retained fraction to estimate growth; we do not assume that all earnings are reinvested.

$$b = 1 - 0.35 = 0.65, \qquad g = b \times ROE = 0.65 \times 0.12 = 7.8\%$$

Step 2. Find the current dividend first.

Now take the shareholder's cash payment out of current earnings.
EPS is earnings per share, while DPS is the part of those earnings actually distributed per share.

$$DPS_0 = EPS_0 \times \text{payout} = 8 \times 0.35 = \$2.80$$

Step 3. Move that dividend forward one year.

The question asks for next year's dividend, so there is one more step.
Grow today's dividend by the rate we just calculated.
Keep the extra decimals here if this dividend will feed into a valuation.

$$DPS_1 = DPS_0(1+g) = 2.80 \times 1.078 = \boxed{\$3.0184}$$

2. Required return with CAPM

Answer A. 10.0%

Step 1. Read the market input carefully.

The first thing I would check is how the market input is described.
Here we are given the expected market return.
We need to remove the risk-free rate to get the compensation for taking market risk.

$$ERP = E[R_M] - r_f = 9\% - 4\% = 5\%$$

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Step 2. Scale the premium by beta.

Next, beta tells us how much exposure this stock has to market risk.
A beta of 1.2 means we apply 1.2 times the market premium, not 1.2 times the full market return.

$$\beta \times ERP = 1.2 \times 5\% = 6\%$$

Step 3. Add compensation for waiting.

Finally, add the risk-free rate back.
Investors need compensation for waiting plus compensation for bearing the stock's market risk.
That gives us the required return to discount equity cash flows.

$$k = r_f + \beta \times ERP = 4\% + 6\% = \boxed{10\%}$$

3. One-stage DDM

Answer C. $41.60

Step 1. Check that the model has a sensible denominator.

Before putting numbers into Gordon, check the assumptions.
We have constant perpetual growth, and the required return exceeds growth.
If growth were at least as high as the required return, this formula would not give a valid finite valuation.

$$k = 12\% > g = 4\%$$

Step 2. Use next year's cash flow.

The dividend in the question was just paid.
It is not the next cash flow a new buyer will receive.
We need to move it forward one year before using the formula.

$$DPS_1 = DPS_0(1+g) = 3.20 \times 1.04 = \$3.328$$

Step 3. Capitalize that dividend.

Now divide next year's dividend by the difference between the required return and growth.
Enter the rates as decimals.
Because the cash flow is per share, the result is already a price per share.

$$P_0 = \frac{DPS_1}{k-g} = \frac{3.328}{0.12-0.04} = \boxed{\$41.60}$$

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4. The discount rate can move the valuation

Answer D. -28.57%

Step 1. Value the same cash flows at the original rate.

Let's do two valuations of exactly the same dividend stream.
Start with the original required return.
The dividend given is already next year's payment, so we do not grow it again.

$$P_{\text{old}} = \frac{4}{0.09 - 0.04} = \$80.00$$

Step 2. Change only the discount rate.

For the second valuation, change only the required return.
The higher discount rate makes the same future cash flows worth less today.
The growth assumption and dividend remain unchanged.

$$P_{\text{new}} = \frac{4}{0.11 - 0.04} = \$57.1429$$

Step 3. Measure the change relative to the original value.

To report a percentage change, compare the new value with the original value.
The original price belongs in the denominator.
This is a price response to a change in required return, not simply the change in the interest-rate input.

$$\%\Delta P = \left(\frac{57.1429}{80} - 1\right) \times 100 = \boxed{-28.57\%}$$

5. Put a two-stage DDM together

Answer C. $24.74

Step 1. Build the value at the transition date.

The explicit forecast ends in year 2.
That means the first dividend in the stable-growth stage arrives in year 3.
Grow the year-2 dividend once, then use Gordon to calculate the value at the end of year 2.

$$DPS_3 = 1.80 \times 1.03 = \$1.854$$

$$TV_2 = \frac{DPS_3}{k - g_s} = \frac{1.854}{0.10 - 0.03} = \$26.485714$$

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Step 2. Keep the explicit dividend and terminal value together at year 2.

Keep the dates straight.
The year-1 dividend gets discounted once.
The year-2 dividend and the terminal value are both measured at the end of year 2, so they get discounted twice.

$$P_0 = \frac{DPS_1}{1+k} + \frac{DPS_2 + TV_2}{(1+k)^2}$$

$$P_0 = \frac{1.50}{1.10} + \frac{1.80 + 26.485714}{1.10^2}$$

Step 3. Discount and add.

Now add the present values.
The terminal value includes dividends from year 3 onward, so it does not replace the year-2 dividend.
Including both is necessary; it is not double counting.

$$P_0 = 1.363636 + 1.487603 + 21.888194 = \boxed{\$24.74}$$